

FINAL TESTING AND VALIDATION PLAN

Project ID: SIH26034
Product: NyayaDrishti-LM
Audience: Entire Engineering Team & Quality Assurance
Status: FROZEN

1. Multi-Tier Testing Strategy Overview

The testing methodology ensures that every statutory rule, optical transformation, and evidentiary hash chain is validated under both simulated mathematical conditions and noisy real-world retail conditions.

THE 4-TIER TEST PYRAMID		
TIER 1: UNIT TESTS INTEGRATION PIPELINE - Deterministic regex syntax session state - Rule AST boolean logic rejection tests - USP floating-point math token linking - Merkle DAG hash propagation latency budget	TIER 2: METROLOGY CALIBRATION - ArUco planar homography vs vernier caliper ground truth - Sub-pixel font height error - Target: MAE <= 0.15 mm	TIER 3: - Multi-panel - Quality gate - OCR polygon - End-to-end
TIER 4: REAL-WORLD PILOT ACCEPTANCE TESTS (50 Physical FMCG SKUs) 50 real retail packages measured with digital vernier calipers (+/- 0.02 mm) - Evaluating false positives (FPR) and false negatives (FNR) on market inventory		

2. Comprehensive Test Suite Specification

Test Suite ID	Subsystem Tested	Test Scenario Description	Test Input Fixture	Expected Deterministic Output	Owner
TS-UNIT-01	Banned Units Parser	Package label contains Net Wt: 500 gms	String "Net Wt: 500 gms"	VIOLATION_FLAG: Section 11 / Rule 12 Banned Unit 'gms'	M3
TS-UNIT-02	Banned Units Parser	Package label contains Volume: 750 ML	String "Volume: 750 ML"	VIOLATION_FLAG: Section 11 / Rule 12 Banned Unit 'ML'	M3
TS-UNIT-03	Valid Units Parser	Package label contains Net Quantity: 500 g	String "Net Quantity: 500 g"	PASS: Valid SI unit 'g'	M3
TS-UNIT-04	USP Math Validator	200 g pack, MRP ₹80, declared USP ₹0.40/g	Qty: 200, MRP: 80, USP: 0.40	PASS:	(0.40 * 200) - 80
TS-UNIT-05	USP Math Validator	400 g pack, MRP ₹200, declared USP ₹0.60/g	Qty: 400, MRP: 200, USP: 0.60	VIOLATION_FLAG:	(0.60 * 400) - 200
TS-UNIT-06	Table-I Font Schedule	PDP Area 450 cm ² (Row 3), measured font 2.8 mm	PDP: 450, Height: 2.8, Req: 2.5	PASS: 2.8 mm >= 2.5 mm Table-I Row 3	M4
TS-UNIT-07	Table-I Font Schedule	PDP Area 450 cm ² (Row 3), measured font 1.8 mm	PDP: 450, Height: 1.8, Req: 2.5	VIOLATION_FLAG: Deficit 0.7 mm citing Row 3	M4
TS-UNIT-	Table-I Row 5 Blown	PDP Area 2800 cm ² ,	PDP: 2800,	PASS: 6.2 mm >= 6.0 mm (Row 5	M4

Test Suite ID	Subsystem Tested	Test Scenario Description	Test Input Fixture	Expected Deterministic Output	Owner
08		Blown container, font 6.2 mm	Blown, Height: 6.2, Req: 6.0	verified)	
TS-UNIT-09	Consumer Care Check	Declaration has Tel, Address, Contact, but no email	Consumer care text block	VIOLATION_FLAG: Rule 6(1)(g) Missing Email Address	M3
TS-UNIT-10	Temporal Epoch Router	Package manufactured 10/2021 (pre-USP amendment)	Mfg Date: 2021-10-15	ROUTE: EPOCH_2017_GSR_629 (Skip Rule 6(1)(f))	M4
TS-UNIT-11	Temporal Epoch Router	Package manufactured 03/2023 (post-USP amendment)	Mfg Date: 2023-03-20	ROUTE: EPOCH_2021_GSR_779 (Enforce Rule 6(1)(f))	M4
TS-UNIT-12	E-Commerce Engine	Online listing missing Manufacturing Date	E-com listing payload	PASS: Rule 6(10) Statutorily Exempts Mfg Date	M4
TS-UNIT-13	E-Commerce Engine	Online listing missing Country of Origin	E-com listing payload	VIOLATION_FLAG: Rule 6(10) / Rule 6(1)(aa)	M4
TS-CALIB-01	ArUco Homography	Planar target tilted 25° with known 20 mm marker	Synthetic tilted render	Scale factor S derived within ± 0.05 mm/px	M1

Test Suite ID	Subsystem Tested	Test Scenario Description	Test Input Fixture	Expected Deterministic Output	Owner
TS-CALIB-02	Font Caliper Parity	Rectified image of physical pack with caliper baseline	Physical pack crop (caliper: 2.45 mm)	Predicted height 2.45 ± 0.12 mm (MAE ≤ 0.15 mm)	M1
TS-OPTIC-01	Quality Gate Blur	Blurred photo caused by camera movement	Laplacian variance $\sigma^2 = 32.4$	UNABLE_TO_VERIFY (IMAGE_BLURRED) with retake prompt	M1
TS-OPTIC-02	Quality Gate Glare	Specular reflection bloom over MRP text	HSV mask ($V > 245, S < 15$) on text crop	UNABLE_TO_VERIFY (SPECULAR_GLARE) with tilt prompt	M1
TS-EVID-01	Merkle DAG Integrity	Full inspection session hashed into DAG	Node payloads from Stage 1 to 6	Merkle Root matches cryptographic recalculation	M5
TS-EVID-02	Tamper Detection	Manually alter 1 character in extracted MRP in JSON	Corrupted JSON payload	Merkle verification fails; flag RECORD_TAMPERED	M5
TS-SYS-01	Offline Execution	Disconnect all network interfaces (airplane mode)	Full physical pack inspection	System executes 100% locally; renders PDF memo	M6

3. Quantitative Acceptance Benchmarks

1. **Optical Metric Accuracy:** Mean Absolute Error (MAE) ≤ 0.15 mm on planar packaging at distances between 15 cm and 25 cm.
2. **Text Recognition Accuracy:** Character Error Rate (CER) $\leq 3.0\%$ on clean printed packaging text; CER $\leq 7.0\%$ on curved or glossy foil crops.
3. **Statutory Extraction F1-Score:** Macro F1 ≥ 0.95 across mandatory fields (MRP, Net Qty, Dates, Origin, Consumer Care).
4. **False Positive Violation Rate (FPR):** $\leq 2.0\%$ on compliant packs (zero false accusations against lawful brands).
5. **False Negative Violation Rate (FNR):** $\leq 1.0\%$ on unambiguous statutory violations (prohibited units, missing origin, USP math).
6. **End-to-End Latency Budget:** < 1.5 seconds total pipeline execution time on a commodity 4-core Intel Core i5 / AMD Ryzen 5 CPU without discrete GPU.